

ELITE MATHEMATICS

Pearson Edexcel International A Level

Pure 2

Adaptive Classified

Chapter 08 · Modelling with Sequences and Series

Questions

Student book · questions with writing space

- 35 pages
- Past papers up to January 2026

PURE 2 | TOPIC 08

Modelling with Sequences & Series

Your classified route

10

QUESTIONS

83

MARKS

4

FAMILIES

5

VARIANTS

DR ESLAM AHMED

CLASSIFIED BY IDEA

Move from recognition to method, then use the working space to complete the solution.

TOPIC 08 | FAMILY 01

Model Fitting

WHAT THIS FAMILY ASKS

Recognise the model fitting demand and complete the required reasoning in a clear, exam-ready order.

VARIANT ROUTE

V01 core family move

2 classified questions

17 total marks

9. A scientist is using carbon-14 dating to determine the age of some wooden items.

The equation for carbon-14 dating an item is given by

$$N = k\lambda^t$$

where

- N grams is the amount of carbon-14 **currently** present in the item
- k grams was the **initial** amount of carbon-14 present in the item
- t is the number of years since the item was made
- λ is a constant, with $0 < \lambda < 1$

- (a) Sketch the graph of N against t for $k = 1$

(2)

Adaptive unit 1 · WMA12-01_2022_MayJune_Q09

DR ESAM AHMED

9. A scientist is using carbon-14 dating to determine the age of some wooden items.

The equation for carbon-14 dating an item is given by

$$N = k\lambda^t$$

where

- N grams is the amount of carbon-14 **currently** present in the item
- k grams was the **initial** amount of carbon-14 present in the item
- t is the number of years since the item was made
- λ is a constant, with $0 < \lambda < 1$

Given that it takes 5700 years for the amount of carbon-14 to reduce to half its initial value,

(b) show that the value of the constant λ is 0.999878 to 6 decimal places.

(2)

DR ESAM AHMED

9. A scientist is using carbon-14 dating to determine the age of some wooden items.

The equation for carbon-14 dating an item is given by

$$N = k\lambda^t$$

where

- N grams is the amount of carbon-14 **currently** present in the item
- k grams was the **initial** amount of carbon-14 present in the item
- t is the number of years since the item was made
- λ is a constant, with $0 < \lambda < 1$

Given that Item A

- is known to have had 15 grams of carbon-14 present initially
- is thought to be 3250 years old

- (c) calculate, to 3 significant figures, how much carbon-14 the equation predicts is currently in Item A .

(2)

Adaptive unit 3 · WMA12-01_2022_MayJune_Q09

DR ESAM AHMED

9. A scientist is using carbon-14 dating to determine the age of some wooden items.

The equation for carbon-14 dating an item is given by

$$N = k\lambda^t$$

where

- N grams is the amount of carbon-14 **currently** present in the item
- k grams was the **initial** amount of carbon-14 present in the item
- t is the number of years since the item was made
- λ is a constant, with $0 < \lambda < 1$

Item B is known to have initially had 25 grams of carbon-14 present, but only 18 grams now remain.

(d) Use algebra to calculate the age of Item B to the nearest 100 years.

(3)

DR ESAM AHMED

10. **In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.**

The number of dormice and the number of voles on an island are being monitored.
Initially there are 2000 dormice on the island.

A model predicts that the number of dormice will increase by 3% each year, so that the numbers of dormice on the island at the end of each year form a geometric sequence.

- (a) Find, according to the model, the number of dormice on the island 6 years after monitoring began. Give your answer to 3 significant figures.

(2)

Adaptive unit 1 · WMA12-01_2024_MayJune_Q10

DR ESAM AHMED

10.

In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

The number of dormice and the number of voles on an island are being monitored.

Initially there are 2000 dormice on the island.

A model predicts that the number of dormice will increase by 3% each year, so that the numbers of dormice on the island at the end of each year form a geometric sequence.

The number of voles on the island is being monitored over the same period of time.

Given that

- 4 years after monitoring began there were 3690 voles on the island
- 7 years after monitoring began there were 3470 voles on the island
- the number of voles on the island at the end of each year is modelled as a geometric sequence

(b) find the equation of this model in the form

$$N = ab^t$$

where N is the number of voles, t years after monitoring began and a and b are constants. Give the value of a and the value of b to 2 significant figures.

(3)

10. **In this question you must show all stages of your working.**
 Solutions relying entirely on calculator technology are not acceptable.

The number of dormice and the number of voles on an island are being monitored.

Initially there are 2000 dormice on the island.

A model predicts that the number of dormice will increase by 3% each year, so that the numbers of dormice on the island at the end of each year form a geometric sequence.

When $t = T$, the number of dormice on the island is equal to the number of voles on the island.

(c) Find, according to the models, the value of T , giving your answer to one decimal place.

(3)

Adaptive unit 3 · WMA12-01_2024_MayJune_Q10

DR ESAM AHMED

TOPIC 08 | FAMILY 02

Interpretation

WHAT THIS FAMILY ASKS

Recognise the interpretation demand and complete the required reasoning in a clear, exam-ready order.

VARIANT ROUTE

V01 For 'at least' or 'exceeds', test the boundary integer.

V02 If the model starts at $n=0$, translate n into calendar time.

3 classified questions

24 total marks

5. **In this question you must show all stages of your working.**
 Solutions relying on calculator technology are not acceptable.

A colony of bees is being studied.

The number of bees in the colony at the start of the study was 30 000

Three years after the start of the study, the number of bees in the colony is 34 000

A model predicts that the number of bees in the colony will increase by $p\%$ each year, so that the number of bees in the colony at the end of each year of study forms a geometric sequence.

Assuming the model,

(a) find the value of p , giving your answer to 2 decimal places.

(3)

Adaptive unit 1 · WMA12-01_2020_Jan_Q05

DR ESAM AHMED

5. In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.

A colony of bees is being studied.
The number of bees in the colony at the start of the study was 30 000
Three years after the start of the study, the number of bees in the colony is 34 000
A model predicts that the number of bees in the colony will increase by $p\%$ each year, so that the number of bees in the colony at the end of each year of study forms a geometric sequence.
Assuming the model,

According to the model, at the end of N years of study the number of bees in the colony exceeds 75 000

(b) Find, showing all steps in your working, the smallest integer value of N . (5)

Adaptive unit 2 · WMA12-01_2020_Jan_Q05

DR ESAM AHMED

8. A metal post is repeatedly hit in order to drive it into the ground.

Given that

- on the 1st hit, the post is driven 100 mm into the ground
- on the 2nd hit, the post is driven an **additional** 98 mm into the ground
- on the 3rd hit, the post is driven an **additional** 96 mm into the ground
- the **additional** distances the post travels on each subsequent hit form an arithmetic sequence

(a) show that the post is driven an **additional** 62 mm into the ground with the 20th hit.

(1)

Adaptive unit 1 · WMA12-01_2022_Jan_Q08

DR ESAM AHMED

8. A metal post is repeatedly hit in order to drive it into the ground.

Given that

- on the 1st hit, the post is driven 100 mm into the ground
- on the 2nd hit, the post is driven an **additional** 98 mm into the ground
- on the 3rd hit, the post is driven an **additional** 96 mm into the ground
- the **additional** distances the post travels on each subsequent hit form an arithmetic sequence

(b) Find the **total distance** that the post has been driven into the ground after 20 hits.

(2)

Adaptive unit 2 · WMA12-01_2022_Jan_Q08

DR ESAM AHMED

8. A metal post is repeatedly hit in order to drive it into the ground.

Given that

- on the 1st hit, the post is driven 100 mm into the ground
- on the 2nd hit, the post is driven an **additional** 98 mm into the ground
- on the 3rd hit, the post is driven an **additional** 96 mm into the ground
- the **additional** distances the post travels on each subsequent hit form an arithmetic sequence

Given that for each subsequent hit after the 20th hit

- the **additional** distances the post travels form a geometric sequence with common ratio r
- on the 22nd hit, the post is driven an **additional** 60 mm into the ground

(c) find the value of r , giving your answer to 3 decimal places.

(2)

Adaptive unit 3 · WMA12-01_2022_Jan_Q08

DR ESAM AHMED

8. A metal post is repeatedly hit in order to drive it into the ground.

Given that

- on the 1st hit, the post is driven 100mm into the ground
- on the 2nd hit, the post is driven an **additional** 98 mm into the ground
- on the 3rd hit, the post is driven an **additional** 96 mm into the ground
- the **additional** distances the post travels on each subsequent hit form an arithmetic sequence

After a total of N hits, the post will have been driven more than 3 m into the ground.

(d) Find, showing all steps in your working, the smallest possible value of N .

(4)

Adaptive unit 4 · WMA12-01_2022_Jan_Q08

DR ESAM AHMED

1. Adina is saving money to buy a new computer. She saves £5 in week 1, £5.25 in week 2, £5.50 in week 3 and so on until she has enough money, in total, to buy the computer.

She decides to model her savings using either an arithmetic series or a geometric series.

Using the information given,

- (a) (i) state with a reason whether an arithmetic series or a geometric series should be used,
- (ii) write down an expression, in terms of n , for the amount, in pounds (£), saved in week n .

(3)

Adaptive unit 1 · WMA12-01_2021_MayJune_Q01

DR ESAM AHMED

1. Adina is saving money to buy a new computer. She saves £5 in week 1, £5.25 in week 2, £5.50 in week 3 and so on until she has enough money, in total, to buy the computer.
- She decides to model her savings using either an arithmetic series or a geometric series.
- Using the information given,

Given that the computer Adina wants to buy costs £350

- (b) find the number of weeks it will take for Adina to save enough money to buy the computer.

(4)

Adaptive unit 2 · WMA12-01_2021_MayJune_Q01

DR ESAM AHMED

TOPIC 08 | FAMILY 03

Limiting Models

WHAT THIS FAMILY ASKS

Recognise the limiting models demand and complete the required reasoning in a clear, exam-ready order.

VARIANT ROUTE

V01 core family move

2 classified questions

21 total marks

10. In this question you must show detailed reasoning.

Owen wants to train for 12 weeks in preparation for running a marathon.

During the 12-week period he will run every Sunday and every Wednesday.

- On Sunday in week 1 he will run 15 km
- On Sunday in week 12 he will run 37 km

He considers two different 12-week training plans.

In training plan *A*, he will increase the distance he runs each Sunday by the same amount.

(a) Calculate the distance he will run on Sunday in week 5 under training plan *A*.

(3)

Adaptive unit 1 · WMA12-01_2021_Jan_Q10

DR ESAM AHMED

10. In this question you must show detailed reasoning.

Owen wants to train for 12 weeks in preparation for running a marathon.

During the 12-week period he will run every Sunday and every Wednesday.

- On Sunday in week 1 he will run 15 km
- On Sunday in week 12 he will run 37 km

He considers two different 12-week training plans.

In training plan *A*, he will increase the distance he runs each Sunday by the same amount.

In training plan *B*, he will increase the distance he runs each Sunday by the same percentage.

- (b) Calculate the distance he will run on Sunday in week 5 under training plan *B*.
Give your answer in km to one decimal place.

(3)

DR ESAM AHMED

10. In this question you must show detailed reasoning.

Owen wants to train for 12 weeks in preparation for running a marathon.

During the 12-week period he will run every Sunday and every Wednesday.

- On Sunday in week 1 he will run 15 km
- On Sunday in week 12 he will run 37 km

He considers two different 12-week training plans.

In training plan A , he will increase the distance he runs each Sunday by the same amount.

Owen will also run a fixed distance, x km, each Wednesday over the 12-week period.

Given that

- x is an integer
- the total distance that Owen will run on Sundays and Wednesdays over the 12 weeks will not exceed 360 km

- (c) (i) find the maximum value of x , if he uses training plan A ,
- (ii) find the maximum value of x , if he uses training plan B .

(5)

Adaptive unit 3 · WMA12-01_2021_Jan_Q10

9. A company mines lithium.
- in year 1, the company mined 12 400 tonnes of lithium
 - in year 15, the company is predicted to mine 7 500 tonnes of lithium

Model A assumes that the mass of lithium the company mines will decrease by the same amount each year.

According to model A ,

- (a) find the mass of lithium the company will mine in year 5

(3)

Adaptive unit 1 · WMA12-01_2026_Jan_Q09

DR ESAM AHMED

9. A company mines lithium.
- in year 1, the company mined 12 400 tonnes of lithium
 - in year 15, the company is predicted to mine 7 500 tonnes of lithium

Model *A* assumes that the mass of lithium the company mines will decrease by the same amount each year.

According to model *A*,

(b) calculate the **total** mass of lithium the company will mine from year 1 to year 15 inclusive.

(2)

DR ESAM AHMED

9. A company mines lithium.
- in year 1, the company mined 12 400 tonnes of lithium
 - in year 15, the company is predicted to mine 7 500 tonnes of lithium

Model *A* assumes that the mass of lithium the company mines will decrease by the same amount each year.

According to model *A*,

Model *B* assumes that the mass of lithium the company mines will decrease by the same percentage each year.

- (c) Find, according to model *B*, the mass of lithium the company will mine in year 10.
Give the answer to the nearest 10 tonnes.

(3)

DR ESAM AHMED

9. A company mines lithium.
- in year 1, the company mined 12 400 tonnes of lithium
 - in year 15, the company is predicted to mine 7 500 tonnes of lithium

Model *A* assumes that the mass of lithium the company mines will decrease by the same amount each year.

According to model *A*,

According to model *B*, there is a limit to the **total** mass of lithium that can be mined.

(d) Calculate the value of this limit. Give the answer to the nearest 100 tonnes.

(2)

DR ESAM AHMED

TOPIC 08 | FAMILY 04

Model Setup

WHAT THIS FAMILY ASKS

Recognise the model setup demand and complete the required reasoning in a clear, exam-ready order.

VARIANT ROUTE

V01 core family move

3 classified questions

21 total marks

2. The adult population of a town at the start of 2019 is 25 000

A model predicts that the adult population will increase by 2% each year, so that the number of adults in the population at the start of each year following 2019 will form a geometric sequence.

(a) Find, according to the model, the adult population of the town at the start of 2032 (3)

Adaptive unit 1 · WMA12-01_2019_Oct_Q02

DR ESAM AHMED

- A model predicts that the adult population will increase by 2% each year, so that the number of adults in the population at the start of each year following 2019 will form a geometric sequence.

(b) Find, according to these models, the total amount of money that would be given to local charity by the adult population of the town from 2019 to 2032 inclusive. Give your answer to the nearest £1 000

(3)

7. Wheat is grown on a farm.
- In year 1, the farm produced 300 tonnes of wheat.
 - In year 12, the farm is predicted to produce 4 000 tonnes of wheat.

Model *A* assumes that the amount of wheat produced on the farm will increase by the same amount each year.

- (a) Using model *A*, find the amount of wheat produced on the farm in year 4.
Give your answer to the nearest 10 tonnes.

(3)

Adaptive unit 1 · WMA12-01_2024_Jan_Q07

DR ESAM AHMED

7. Wheat is grown on a farm.
- In year 1, the farm produced 300 tonnes of wheat.
 - In year 12, the farm is predicted to produce 4000 tonnes of wheat.

Model *A* assumes that the amount of wheat produced on the farm will increase by the same amount each year.

Model *B* assumes that the amount of wheat produced on the farm will increase by the same percentage each year.

- (b) Using model *B*, find the amount of wheat produced on the farm in year 2.
Give your answer to the nearest 10 tonnes.

(3)

Adaptive unit 2 · WMA12-01_2024_Jan_Q07

DR ESAM AHMED

- (c) Calculate, according to the two models, the difference between the total amounts of wheat predicted to be produced on the farm from year 1 to year 12 inclusive. Give your answer to the nearest 10 tonnes.

Adaptive unit 3 · WMA12-01_2024_Jan_Q07

7. Jem pays money into a savings scheme, A , over a period of 300 months.

Jem pays £20 into scheme A in month 1, £20.50 in month 2, £21 in month 3 and so on, so that the amounts Jem pays each month form an arithmetic sequence.

(a) Show that Jem pays £69.50 into scheme A in month 100

(1)

Adaptive unit 1 · WMA12-01_2024_Oct_Q07

DR ESAM AHMED

7. Jem pays money into a savings scheme, A , over a period of 300 months.

Jem pays £20 into scheme A in month 1, £20.50 in month 2, £21 in month 3 and so on, so that the amounts Jem pays each month form an arithmetic sequence.

(b) Find the **total** amount that Jem pays into scheme A over the period of 300 months.

(2)

Adaptive unit 2 · WMA12-01_2024_Oct_Q07

DR ESAM AHMED

7. Jem pays money into a savings scheme, A , over a period of 300 months.

Jem pays £20 into scheme A in month 1, £20.50 in month 2, £21 in month 3 and so on, so that the amounts Jem pays each month form an arithmetic sequence.

Kim pays money into a different savings scheme, B , over the same period of 300 months.

In a model, the amounts Kim pays into scheme B increase by the same percentage each month, so that the amounts Kim pays each month form a geometric sequence.

Given that Kim pays

- £20 into scheme B in month 1
 - £250 into scheme B in month 300
- (c) use the model to calculate, to the nearest £10, the difference between the total amount paid into scheme A and the total amount paid into scheme B over the period of 300 months.

(3)

Adaptive unit 3 · WMA12-01_2024_Oct_Q07

DR ESAM AHMED